SUN YANHE

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M.Sc. candidate, Surveying & Geospatial Engineering, Southwest Jiaotong University | Graduating Jun 2027 | Target: GIS / Digital Twin Front-End Engineer

EDUCATION

Southwest Jiaotong University — M.Sc., Surveying, Mapping and Geospatial Information Engineering Sep 2024 – Jun 2027

Admitted by examination waiver. 2nd Prize, National Virtual Geographic Environment Innovation Competition; Sichuan College Student Comprehensive Quality Certificate (Grade A); 2nd-Class Academic Scholarship; Mingcheng Award.

Chongqing Jiaotong University — B.Eng., Surveying and Mapping Engineering Sep 2020 – Jun 2024

Recommended for M.Sc. by examination waiver. National Encouragement Scholarship (×2); Outstanding Graduate (Chongqing Municipality); Three-Good Student; Outstanding Youth League Member; Campus Football Best XI (2021–22).

2nd Prize (National), UAV Aerial Survey Virtual Simulation Contest; 2nd Prize (SW Region), National Marine Vehicle Design & Construction Competition; 3rd Prize (Municipal), China College Computer Design Competition.

PROFESSIONAL EXPERIENCE

Sichuan Transportation Survey & Design Research Institute Co., Ltd. — Web Front-End Developer, Digital & Intelligent Transportation Branch Jun 2026 – Present

Delivered and iterated multiple government / enterprise GIS and web systems: data processing, module development, architecture optimisation, award applications and proposal writing.

- **Communication Conduit Management Platform:** built the province-wide admin backend covering contracts, conduit asset management and role-based permission assignment; now in production.
- **UAV Panoramic Service Platform:** developed the panorama editor with viewpoint control and hotspot annotation, presenting target points intuitively on immersive imagery.
- **UAV Inspection Digital Twin Platform:** implemented front-end visualisation of inspection data (airports, work sections, labels, flight routes, aerial results, aircraft records) and label-based annotation of aerial images, closing the loop from annotated samples to auto-generated daily inspection reports.
- **Panzhihua–Yanjin Expressway GIS Visualisation Platform:** fused design data (bridges, tunnels, interchanges, chainages, work sections) with terrain, DOM, BIM models, panoramas and UAV media on a 3D basemap; built an earthwork tool computing cut-and-fill volumes from polygons drawn on the map.
- **Aba Prefecture Highway Command & Dispatch Platform:** linked hazards, congestion and work points/sections with maintenance stations in an interactive map view for the Aba Highway Bureau.

Sichuan Wanyichangzhuo Information Technology Co., Ltd. — Surveying Intern (Field & Office) Jul 2023 – Sep 2023

Participated in stake-out setting and topographic mapping; mastered the end-to-end UAV aerial survey workflow from field acquisition to office deliverables.

RESEARCH EXPERIENCE

National Invention Patent (Lead Inventor) — VR Panoramic Flood Analysis Combining Knowledge Graph and Vision LLM Jul 2025 – Nov 2025

Built a knowledge-graph-constrained vertical vision LLM, establishing an end-to-end pipeline from image understanding to recognition and reasoning for accurate flood damage assessment.

China State Railway Group, Major R&D Program (Core Researcher) — Smart Construction along Chengdu–Chongqing Middle Line HSR May 2025 – Feb 2026

Designed the intelligent judging algorithm for box-girder pedestal retrofitting and the production-plan output logic; modelled dynamic scheduling from construction plan, pedestal status and erection demand, and integrated it into the web control system for a low-staff precast yard.

Peking University Advanced Technology Institute (Core Researcher) — 3D Urban Scene Fusion from Satellite Imagery Oct 2024 – Feb 2025

Led development of a 3D urban reconstruction prototype fusing satellite remote sensing, UAV oblique photogrammetry and ubiquitous sensing data; implemented Cesium-based model loading, visualisation and scene roaming; contributed to technical validation and the closing report.

China State Railway Group, Key R&D Project (Core Researcher) — Flood Simulation and Emergency Response for Underground Stations under Rainstorms Jul 2024 – Jan 2025

Developed a digital-twin prototype for rainfall-induced flooding at underground stations; matched risk assessment with emergency drill scenarios and verified at pilot sites.

National Key R&D Program (Participant) — Portable On-Site Disaster Detection and Assessment Platform Jan 2025 – Present

Rainstorm flood simulation sub-task: built a flood disaster knowledge graph from multi-source data, integrating numerical models with geo-spatiotemporal semantic rules for “scenario – deduction – response” analysis.

LEADERSHIP & ACTIVITIES

Head of Sports & Culture Dept., Graduate Union, School of Geoscience and Engineering, SWJTU 2024 – 2025

Planned and ran the freshman basketball tournament; handled publicity, photography and cross-department coordination.

Vice President, Chongqing Jiaotong University Football Association 2021 – 2023

Organised the Jiaobei Cup and Super League (500+ registrants); ran the official WeChat and Douyin accounts to 1,700+ followers, single video 13K+ likes.

Captain & Midfielder, School of Civil Engineering Football Team 2021 – 2022

Led the team to 3rd place in the campus futsal tournament and 2nd in the Jiaobei Cup; selected to Campus Football Best XI.

SKILLS & CERTIFICATIONS

Front-End & GIS: React, Vue, Cesium, JavaScript, Python; digital twin and 3D visualisation.

Other: Photoshop, Office suite, LLM application, data processing, project application and technical writing.

Certifications: Driving Licence; National Grade-3 Football Referee. Interests: football, badminton, hiking, mountaineering.